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REVIEW ARTICLE

A Systematic Review of Concept Drift Approaches for Anomaly Detection in IoT Networks

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Abstract
Internet of Things (IoT) is a networking solution for providing smart services across many domains. Ever since the emergence of IoT, providing adequate security for its ecosystem has been a persistent challenge, thereby making IoT networks and infrastructures highly susceptible to various network anomalies and cyber-attacks. Despite the effectiveness of machine learning based intrusion detection systems (IDSs) in detecting network anomalies and cyber-attacks, they are vulnerable to concept drift phenomenon that degrades the accuracy of the model over time. Over the recent decade, several approaches have been proposed to address the effects of concept drift on anomaly detection methods. The aim of this paper is to provide a systematic review of concept drift approaches for anomaly detection in IoT networks. Thus, it sought to examine how stream learning has improved detection of cyber-attacks in real-time by reviewing recent studies that incorporate concept drift techniques for anomaly detection in IoT environment. The review process includes an extensive literature search of relevant scientific studies that satisfy the pre-defined inclusion criteria, follow by the analysis and thorough synthesis of findings from the selected literatures. The study examines and identifies the trending concept drift approaches including the detection methods and adaptation strategies in IoT environment. It also reveals the common datasets and evaluation methods employed in the field. The review implication for future research direction is highlighted in the study, hence the need for further studies to advance the security of IoT systems to achieve the long-time vision of smart computing for automatic service delivery.

Keywords Stream learning · Concept drift · IoT security · Anomaly detection · Cyber-attack detection

Introduction

Internet of Things (IoT) is a network of billions of cyber-physical devices, communicate via the Internet automatically without human intervention. [1, 2]. IoT is becoming an important infrastructure for powering various applications and providing smart services in the areas of healthcare, agriculture, power, transportation, security and surveillance, and environmental planning [3]. Over the last decade, the number IoT applications and users has increased tremendously, thereby making the amount of data generated by IoT devices grow exponentially. The IoT data are generated continuously in streams, a feature that necessitates a real-time data processing and analysis. However, IoT devices are highly vulnerable to security threats and attacks such as Botnet, Phishing, Denial of Service (DoS), spoofing, and Distributed Denial of Service (DDoS). These threats occur as a result of inherent limited resources in the IoT devices, such as power, memory and processors. Unfortunately, these requirements make the manufacturers pay less attentions to implementation of stringent security measures [3, 4]. As the IoT devices gain popularity, the number of attacks on the system is increasing exponentially. These types of network attacks can disrupt access to the various smart services provided by the IoT and capable of affecting the entire Internet ecosystem. Hence, an effective monitoring and cyber-attack detection and mitigation system is required.

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In recent years, researchers are increasingly leveraging the power of data analysis and data mining to identify infected IoT devices that can be exploited by cyber-attackers to cause network intrusion and other attackers. Meanwhile, Intrusion detection system, a widely used security system for monitoring and analyzing network traffic and protect IoT ecosystem from malicious cyber-attacks, employs network traffic data analysis based on two mechanisms, namely signature-based IDS and anomaly-based IDS [5]. The Signature (or misused) based IDS detects compromised devices or malicious attacks by comparing the incoming network packets against the predefined signatures or attack patterns. On the other hand, anomaly IDS is a behavioral based technique that detects malicious attacks by comparing the incoming network traffic against a set of legitimate behaviors, and any deviation from normal activities is regarded as anomaly behavior. Anomaly based solutions perform better than misused based solution as it can detect zero-day attacks [6]. Meanwhile, anomaly-based solutions employ machine learning (ML) or deep learning techniques to analyze and detect anomalous in network traffics [7, 8].

Meanwhile, most of the existing machine learning techniques for anomaly detection systems are facing serious challenges when deployed in IoT set-up. Unlike the traditional networks, IoT network consists of ubiquitous connected devices sharing high volume of data with varying velocity arriving in real-time and continuous manner. More so, IoT data stream have tendency to change over time, thereby making analysis on the previous data instance less effective. The tendency of data stream distribution to change over time is known as concept drift [6]. Thus, the nature of IoT network data corresponds to the characteristics of data stream with high tendency to change over time. With these requirements, it becomes necessary for every modern-day machine learning-based model deployed in the IoT networks to incorporate data stream learning approaches, including concept drift and adaptive learning techniques. These allow ML-models to learn in real-time considering the continuous and non-stationary nature of IoT data. Concept drift learning is a technique that allows machine learning model identifies and adapts to changes in data distribution that could cause degrades in its performance. On the other hand, adaptive learning allows model to continuously learn with respect to the incoming instances of the data stream. These techniques are gaining a widespread adoption in many modern machine learning based systems across many domains such as medical [9], multimedia [10], social networking [11] and network security [12, 13].

Meanwhile, data stream learning is increasingly receiving attention due to its ability to allow classifiers discover hidden knowledge from high volume of data stream which could lead to change in data distribution. This motivation is strengthened by some recent studies [6, 14, 15] that have attributed the failure of some traditional ML based cyber-attacks detection models in IoT environment to their inability to handle concept drift phenomenon in the network traffic data. Most of these models were based on traditional machine learning techniques that employed static data for training against the actual situation in IoT networks, where data samples are generated continuously in streams [16, 17]. Typically, when data drift occurs in anomaly detection, the model becomes outdated and its accuracy performance also becomes degraded. As a result, the model will become ineffective in detecting cyber-attacks. In this case, normal traffics may be taken as attack traffics and vice versa. Hence, the model needs to be retrained in order to adjust to the drift in network traffic so as to maintain its optimal performance. Hence, addressing concept drift is very important for modern machine learning based security solutions because of ever evolving nature of security threats landscape and insufficient labeled data at the deployment. In essence, an effective machine learning based network anomaly detection system must be effective and acclimate to the occurrence of concept drift and detect cyber-attacks in real-time.

Therefore, the aim of this paper is to introduce a survey that reviews the recent studies on anomaly detection techniques for IoT networks with the view of identifying the state-of-art- ML techniques that incorporate data stream learning and concept drift analytics for cyber-attack detection in IoT networks. The objectives of this study is to identify the trending concept drift methods for anomaly detection in IoT environment and their effectiveness; identify the common techniques or strategies employed in recent literature for handling the occurrence of concept drift; determine the prominent evaluation datasets and their constituent cyber-attack types; and determine the common evaluation methods for training and testing concept drift enabled models for IoT networks.

Meanwhile, this study is guided by the following research questions:

1) What are the trending concept drift detection methods for anomaly detection in IoT networks?
2) What are common methods and strategies for adapting anomaly detection models to concept drift?
3) What are the common datasets and prominent cyber-attacks in the datasets adopted for concept drift-based anomaly detection models?
4) What are the common methods for evaluating concept drift-based models?

Furthermore, this study provides a comprehensive guide for Information and Cybersecurity researchers, experts and IoT device manufacturers while selecting concept drift

detection and handling methods for building effective security solutions or devices for detecting cyber-attacks in the IoT environment.

The remaining sections of this study comprise literature review in Sect. 2 where related concepts such as IoT security issues, anomaly detection methods and concept drift learning were presented. Section 3 presents the research methodology utilized in this study, while Sects. 4, 5 and 6 present the results, discussion of findings, conclusions and future research direction respectively.

Literature Review

Internet of Things is an emerging technology that is gaining increasing research attentions from diverse academic communities. The increasing adoption of IoTs as well as the sophistication and dynamic nature of cyber-attacks on IoT networks and devices makes researches on IoT security highly imperative. Therefore, this section presents some related concepts to this subject area, including the concept of IoTs and related security issues, anomaly detection techniques for IoTs, and the concept of concept drift for anomaly detection in IoT networks.

Internet of Things and Security Issues

Internet of Things (IoT) refers to a computing technology consisting of networks of thousands or billions of smart enabled devices that collect and exchange data over the Internet without human intervention [2]. The IoT devices that are popularly called "things" made up of electronic sensors, actuators, Internet connectivity and application programs for providing various autonomous or smart services. Internet of Things technology has enormous benefits and its application is gaining wide adoption in many sectors such as transportation for autonomous vehicles and traffic monitoring and control; healthcare delivery for patient monitoring and quality drug administration; housing and urban planning and development for smart homes, and smart city; power and industries as smart grid and smart industry systems respectively [4]. In essence, the goal of IoT technology and its smart solutions is to improve human lives and be more productive.

Nevertheless, ever since its emergence, providing network security to IoT devices and its ecosystem has been a serious challenge, as it is highly susceptible to various cyber threats and cyber-attacks [18] as a result of their openness and ability to be accessed from remote location through the internet by heterogenous IoT devices with inherent computer resource limitations and weak security mechanisms. These vulnerabilities make IoT devices, networks and infrastructures prone to security threats and attacks such as spying, man-in-the-middle, password cracking, probing, network intrusion, backdoor, cross-site-scripting (XSS), Identity theft, phishing, spamming, and ransomware [19]. In addition, the issues being faced in IoT networks and ecosystem are overwhelming and poses serious concerns to the manufacturers and academia. Razaei [20] highlighted common open security issues in IoT environment to include botnet and distributed denial of service (DDoS), cryptographic algorithms, authentication protocols, access control and governance framework. These issues are so critical and pose security threat to IoT system and the entire Internet [21, 22].

Anomaly Detection Techniques for IoT Security

Anomaly detection is a technique that uncovers unusual behavior from a system, such as a network that deviates from the predicted patterns [23]. Anomaly detection techniques has been deployed in several applications, such as analysis of network, intrusion detection, fraud detection, system error detection, traffic and other monitoring systems. In a network environment such as IoT, anomalies occur as a result of factors sensor failure, network intrusion, internal and/or external attacks [24]. Machine learning is an effective technique for building anomaly detection models in IoT using approaches such as supervised learning, unsupervised learning, or semi-supervised learning. In IoT, the intelligence of the machine learning technique algorithms is used to uncover unusual patterns such as network intrusions and security attacks. Several studies have proposed ML-based models to detect different IoT network anomalies and cyber-attacks, such as [25–28]. However, recent studies [4, 12, 13, 22] have advocated the importance of adopting stream learning that includes concept drift detection and adaptation mechanisms for anomalies detection methods in IoT environments.

Concept Drift Learning for Anomaly Security Detection

Concept drift is a data analytic phenomenon resulted from a change in data distribution over time [29, 30]. It is an emerging research field operating on a principle that statistical characteristic of continuous data, changes over time, and the new data is highly influential than old one in any predictive or data-driven analysis such as machine learning and data mining [31, 32]. Therefore, to accurately classify the next records, concept drift must be handled appropriately. According to [15], there are six approaches of concept drift. These include gradual, sudden, recurring, incremental, blip, and noise. Irrespective of the drift type, concept drift can be grouped into two: real drift and virtual drift.

The real concept drift refers to a situation where there is a conditional change in the target variable while there may exist a distribution change in the input features. On the other hand, virtual concept drift occurs when there is an incomplete representation of data distribution without changing the underlying data distribution [33]. Typically, in a network environment such as IoT, anomaly occurs as a result of external factors, like external attack, and sensor failure [24]. A technique for handling the occurrence of concept drift is highly important for any modern machine learning based security solution, especially in IoT environment where the data stream is non-stationery and cyber-attack landscapes tend to change from time to time.

Concept drift analysis is gaining a widespread adoption in modern machine learning-based models across many domains, such as medical [9], multimedia [10], social networking [11] and network security [12]. The approach has proved to be highly important for consistency and accuracy of anomaly detection models, especially in security domain. IoT network anomaly needs to implement techniques for handling change in data stream due to the dynamic nature of attack landscape and patterns. Most importantly, it is highly important that approaches for handling drift in data distribution is incorporated during the training and testing of the model [34, 35]. Concept drift impacts the performance of anomaly detection model by reducing the accuracy of the model and increases it false positive rate [36]. When concept drift occurs, it makes anomaly detection model becomes outdated and less accurate in detecting intrusion and cyber-attack threats. Thus, with an outdated model, normal traffics may be taken as attack traffics and vice versa [37]. Therefore, incorporating drift detection and handling mechanisms in anomaly detection become imperative as it will not only detect network anomalous effectively, but will also ensure the model retain its optimal performance at all times [35].

Basically, concept drift learners for drift detection and adaptation are usually implemented as active learning approach and passive approach [38]:

- Active learning approach activates it concept drift adaptation strategy explicitly when drift is detected. In most cases, a separate classier is used for drift detection and another one is used for handling the change in data distribution.
- Passive learning approach continuously updates or adjusts the model implicitly over time. This approach is made up of classifiers with an in-built adaptive capability such as ARF [39] and HAT [14].

Concept Drift and Adversarial Perturbation

Apart from concept drift phenomenon, adversarial perturbation is another form of challenge that impacts the performance of machine learning model. Unlike the concept drift that depicts change in data distribution over time, adversarial perturbation is a phenomenon that describes the exploitation of ML model vulnerability to inject noisy or small perturbations to cause changes in input data for the purpose of misleading the model to output wrong prediction [40]. Thus, concept drift is a key characteristic of real-life data distribution of a dynamic environment such as IoT or cyber-physical systems [29, 30]. Although, both phenomena cause changes in data distribution and reduce the performance of anomaly detection models, they are different in term of their goals and targets. While concept drift is largely a natural occurrence in data distribution [24], the adversarial perturbations, such as data poisoning, evasion, inference, model extraction, model reverse, are artificial and a deliberate modification to the training dataset to mislead ML models to produce wrong predictions [40].

In essence, concept drift and adversarial attacks are two different ML vulnerabilities that affect the performance of anomaly detection model. They are similar when it comes to causing changes in data distribution in anomaly detection models. However, in term of occurrence and targets, concept drift does occur naturally because of the dynamic nature of cyber-physical systems such as changes in network usage pattern, user preference, hardware or software failure. The adversarial attacks are caused by attackers with the intension to exploit ML model by changing the input dataset.

More so, the increasing cyber-attacks on cyber-physical systems and infrastructures, such as smart grids, autonomous vehicles, smart transport, and smart health, calls for more cyber-resilience solutions capable of addressing the identification, prevention, detection, mitigation, and recovery issues in IoT ecosystems [41]. More importantly, incorporating concept drift techniques into the cyber-resiliency solutions is becoming necessary as it is capable of enhancing the security solutions by allowing early detection of changes in data distribution that can mislead ML models in real-time.

Meanwhile, the focus of this study is to conduct a review of recent peer review articles that implemented concept drift approaches to enhance the performance anomaly detection models in IoT networks.

Table 1 Articles inclusion criteria

S/*N*	Criteria for inclusion
1.	The study focuses of concept drift approaches, anomaly detection models, machine learning model, Internet of Things
2.	The article was peer reviewed and published in reputable journal and conference
3.	The article written in English
4.	The article could be survey or research paper
5.	The article was published between 2020 and 2023
5.	The article is available online

Table 2 Searched academic databases with article number

S/*N*	Source of Article	URL	No of Articles
1.	Science Direct	http://www.sciencedirect.com/	14
2.	IEEE Xplore	http://www.ieeexplore.ieee.org/	13
3.	Google Scholar	https://scholar.google.com/	20
4.	Web of Science	https://apps.webofknowledge.com/	3
5.	ACM Digital Library	http://dl.acm.org/	3

Research Methodology

This study adopted a research design that includes a systematic review of relevant articles for analysis and synthesis for solving research problems. As a guide, this study followed the Preferred Reporting Items for Systematic Reviews and Meta-analysis for Protocol (PRISMA-P) 2015 guideline outlined in [42]. The guideline originally developed for the analysis of health-related studies, but now widely adopted in many fields of studies to ensure the systematic review is well planned and systematically documented in a concise and transparent manner thereby promoting integrity and accountability of the review articles.

Therefore, this section highlights the strategies employed to search the articles used for this study. It also highlighted the sources of the articles included for analysis, while the key points from the articles are synthesized and summarized in Tables 1, 2, 3, 4, 5, 6 and 7; Figs. 1 and 2.

Data Sources

A systematic review relies heavily on the quality of its data sources where articles used for review are identified and selected. These sources include variety of academic databases and journals that are searched and scrutinized for literatures that are relevant to the field of study. Meanwhile, this study considered reputable academic databases and journals including Science Direct, Google Scholar, IEEE Explore, Springer, Wiley Online Library, Taylor & Francis, and Web of science, as a result of their popularity and quality of journal or conference articles indexed by them.

Search Keywords and Strategy

To ensure relevant articles are retrieved and appropriate search terms are used, this study adopted Kitchenham et al., [43] search strategy for the literature search and sometimes included substrings with the main search strings for the inclusion of all relevant articles and overcome some problems associated with the used databases. During the article search phase, the search terms used as keywords are "concept drift", "anomaly detection", "IoT", "machine learning", and "cyber-attack", "concept drift + anomaly detection + IoT", "Concept drift approach + IoT + cyber-attack".

Inclusion and Exclusion Criteria

In order to avoid bias and ensure each paper from the adopted academic databases and library is given equal chance of being selected, a clear inclusion and exclusion principle was formed and used to identify the articles used for review and synthesis analysis. Similarly, to ensure the study keeps track of the recent developments in term of concept drift techniques and approaches for anomaly detection in IoT environment and remain focus, this study considered only the peer reviewed articles, written in English and published between the year 2020 and 2023. In addition, two recent articles published in 2024 were included in the course of revision of this article. The details of the inclusion criteria are presented in Table 1:

Exclusion from Pooled Article

The original search period for the included articles was May 2023 and all the publication found at the time of the search which led to the pool of 5,235 articles were used for the review and analysis. The screening process to reduce articles from the pool to meet the goal of the study is outlined as follows in Fig. 1:

1. Elimination of article due to unrelated title, duplication from different sources resulted into reduction from 5,235 to 1,048;
2. Elimination of paper after examining the abstract 1,048 articled resulted into reduction to 358;
3. Elimination of article after reading through led to a reduction from 358 to 53.

Thus, Fig. 1 shows the flow of the elimination procedure for screening the articles used in this study.

Table 3 Summary of concept drift approaches for anomaly detection in IoT networks

Article	Focus	Concept Drift Detection Technique	Concept Drift Adaptation/Anomaly detection technique	Detection mechanism	Learning strategy	Dataset	Performance Metrics
[14]	Botnet detection in IoT network	ADWIN	NB, HAT, ARF		Active and passive learning	Self-generated	ARF: 0.9995 accuracy, 0.9990 Kappa statistics (K), and 0.9991 Kappa mean
[2]	Anomaly detection with concept drift detection and adaptation	Sliding window	Adaptive window-based PSO (for drift adaptation), and LightGBM (for anomaly detection)		Active and passive learning	IoTID20 and NSL-KDD	accuracy, precision, recall, F1 measure and average test time
[15]	Improve anomaly detection with neural network and concept drift learning approaches	Stratified sampling technique	Improved Long Short-Term Memory (I-LSTM) optimized with RMSprop optimizer and Softlink activation		Active and passive learning	self-generated smart home emulat-ed environment	
[6]	Improve anomaly detection in dynamic environment using Deep learning	Dynamic sliding window based on residual projection	CNN and LSTM		Active learning	BoT-IoT dataset, SEA Concept and UG-2 C-5D	
[46]	Develop effective anomaly detection system for dynamic environment	Principal component analysis (PCA) and online outlier detector	Online deep neural network based on hedge weighting mechanisms		Active learning	DS2OS	Minimum accuracy of 96.8% and maximum of 100%
[39]	Develop a drift adaptive model for dynamic environment	Adaptive Window (ADWIN)	Adaptive Random Forest (ARF) classifier		Active and passive learning	CICIDS2018 dataset	Overall accuracy (99.5%) and recall (99.8%)
[4]	Improve anomaly detection using ensemble concept drift method	ADWIN, DDM	Ensemble of ARF-ADWIN, ARF-DDM, SRP-ADWIN, and SRP-DDM	PWPAE	Active and passive learning	IoTID20 and CICIDS2017	accuracy, precision, recall, and f1-score, and execution time
[47]	Using concept drift model for anomaly detection in stream network data	K-means based sliding window algorithm	Ensemble of random forest (RF) and logistic regression (LR)	K-means clustering	Active learning	NSL-KDD, CIDDS 2017 and Testbed generated datasets	accuracy of 93%, 98% and 97% was achieved on NSL-KDD, CIDDS and Testbed datasets respectively
[48]	IoT anomaly detection with drift adaptive mechanism	ADWIN, EDDM	Ensemble of ARF-ADWIN, ARF-EDDM, OPA and KNN-ADWIN.	adaptive exponentially weighted average Ensemble (AEWAE).	Active and passive learning	CICIDS2017 and IoTID20	99% accuracy and precision on the two datasets with minimal latency
[49]	Improve ensemble-based anomaly detection method for dynamic environment	-	relevance vector machine (RVM)	overall misclassification probability voting weight value (OMPW)	Passive learning	KDD CUP 99 dataset and CICIDS2017 dataset	98% detection rate on training data and 99% on the incremental sets.

Table 3 (continued)

Article	Focus	Concept Drift Detection Technique	Concept Drift Adaptation/Anomaly detection technique	Detection mechanism	Learning strategy	Dataset	Performance Metrics
[50]	Provide a real-time anomaly detection in IoT	-	Ensemble bagging technique based on deep learning		Passive learning	NSL-KDD, KDD CUP 99 and Kyoto 2006 +	over 99% accuracy on NSL-KDD and KDD CUP 99 datasets and over 98% on Kyoto 2006 + dataset.
[51]	Anomaly detection with concept drift adaptation		Adaptive bagging	internal weight update classifier scheme	Passive learning	KDD CUP 99	
[35]	A concept drift-based Anomaly intrusion detection	HT	Heterogenous ensemble of HT-ARF	Bagging		KDDCup99 dataset	

Table 4 Concept drift detection techniques and category

Publication	Concept Drift Detection Method	Category
[15]	Stratified sampling technique	Sampling-based method
[47]	K-means clustering	
[46]	Principal component analysis (PCA)	
[14, 39]	Adaptive window (ADWIN)	Window-based Method
[2]	Optimized Adaptive and Sliding Windowing (OASW)	
[15]	K-Means-based Sliding Window	
[4]	Drift Detection Method (DDM)	Performance-based Method
[4]	Early Drift Detection Method (EDDM)	

Table 5 Concept drift adaptation techniques and learning strategies

Article	Concept Drift Adaptation Method	Learning Strategy
[14]	Hoeffding Adaptive Tree (HAT)	Adaptive Learning Technique
[15]	I-LSTM	
[19]	P-ResNet	
[46]	Online Deep Neural Network	
[39]	Adaptive Random Forest (ARF)	
[6]	Deep Learning (CNN & LSTM)	
[3]	LightGBM	Ensemble learning Technique
[4]	ARF-ADWIN, ARF-DDM, SRP-ADWIN, SRP-DDM	
[47]	LR + RF + SVM	
[48]	AEWAE	
[49]	OMPW	
[50]	Bagging-based IbEC	
[51]	Adaptive-bagging	
[35]	HT + ARF	

Data Collection and Result Synthesis

Inline with the adopted review guideline, 53 articles were selected and the analysis was performed to extract related information regarding the objectives of this study. For

Table 6 Prominent Cyber-security datasets and attacks for concept drift-based anomaly detection in IoT networks

Author	Dataset Name	Dataset Category	Cyber-Attack Type
[15]	Smart home Dataset	Self-generated cyber-security Dataset	malicious scan, DoS attack, Spying, malicious control, data probing, wrong setting, and malicious operation
[14]	Botnet IoT Dataset	Self-generated cyber-security Dataset	Mirai botnet attack
[3]	IoTID20 and NSL-KDD	Public cyber-security Dataset	IoTID20: DoS, Mirai, MITM, and Scan NSL-KDD: DoS, Remote to Local (R2L), Probe and User to Root (U2R)
[4]	IoTID20 and CICIDS2017	Public cyber-security Dataset	CICIDS2017: Brute Force, Botnet, DoS, Heartbleed, Botnet, Web attack, Infiltration Attack
[6]	Bot-IoT dataset	Public cyber-security Dataset	DoS, Information theft and Probing
[47]	NSL-KDD dataset, the CIDDS-2017 dataset and generated Testbed dataset	Self-generated and Public cyber-security Dataset	Self-generated Testbed dataset: Slowloris, Slow HTTP Body and Slow HTTP Read.
[46]	DS2OS	Public cyber-security Dataset	DS2OS: DoS: Data Probing, Malicious Control, Malicious Operation, Network Scan, Spying, and Wrong Setup
[35, 51]	KDD CUP 99	Public Cyber-security dataset	KDD CUP 99: DoS, U2R, Probe, Others

Table 7 Summary of evaluation methods

Article	Validation methods	Number of Dataset	Accuracy	precision	Recall	F1-score	Execution time and others	Confusi-on matrices	Kappa Statistics
[14]	Prequential	1	✓	-	-	-	-	-	✓
[15]	Hold-out	2	✓	✓	✓	-	-	-	-
[3]	Hold-out and prequential	2	✓	✓	✓	-	-	-	-
[4]	Hold-out and prequential	2	✓	✓	✓	-	✓	-	-
[6]	Hold-out	3	✓	-	-	-	✓	✓	-
[53]	Hold-out	3	-	✓	✓	✓	-		-
[47]	Hold-out	3	✓	✓	✓	✓	-	✓	-
[46]	Hold-out with K-fold cross-validation	1	✓	✓	✓	✓	-	✓	-
[39]	Hold-out	1	✓	✓	✓	✓	✓	✓	-
[48]	Hold-out and prequential	2	✓	✓	✓	✓	✓	✓	-
[49]	Hold-out	3	✓	-	-	-	✓	✓	-
[50]	Hold-out	2	-	✓	✓	✓	-	-	-
[35]	prequential	1	✓	-	-	-	✓	-	-

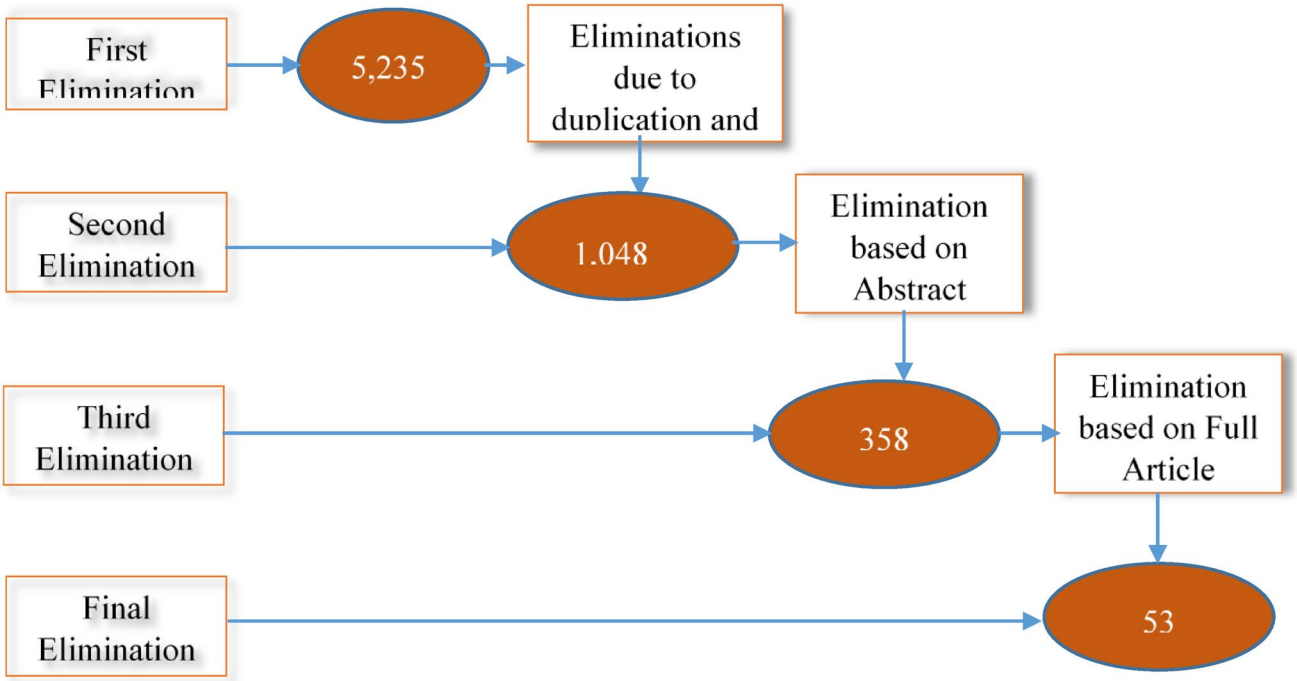


Fig. 1 Articles elimination procedure

proper analysis and synthesis of the papers, the extracted information from the articles were presented in a tabular form and the impressions gained from the reviews were also tabulated and synthesized within the text. In addition to the initial 53 papers, 3 more recent articles between 2024 and 2025 were included for synthesis analysis during periods of peer reviewing and revision. In essence, the next sections present the results of the review papers and some insights of the findings for ease of analyses and synthesis.

Presentation of Review Results

Following the careful selection of articles according to the inclusion criteria as itemized in Sect. 3.3, some experimental based articles with details of their experimental setups including the techniques for effective anomaly detections as well as the mechanisms for identifying and handling concept drift phenomenon were identified. Therefore, the findings from the systematic review are presented in this section thus:

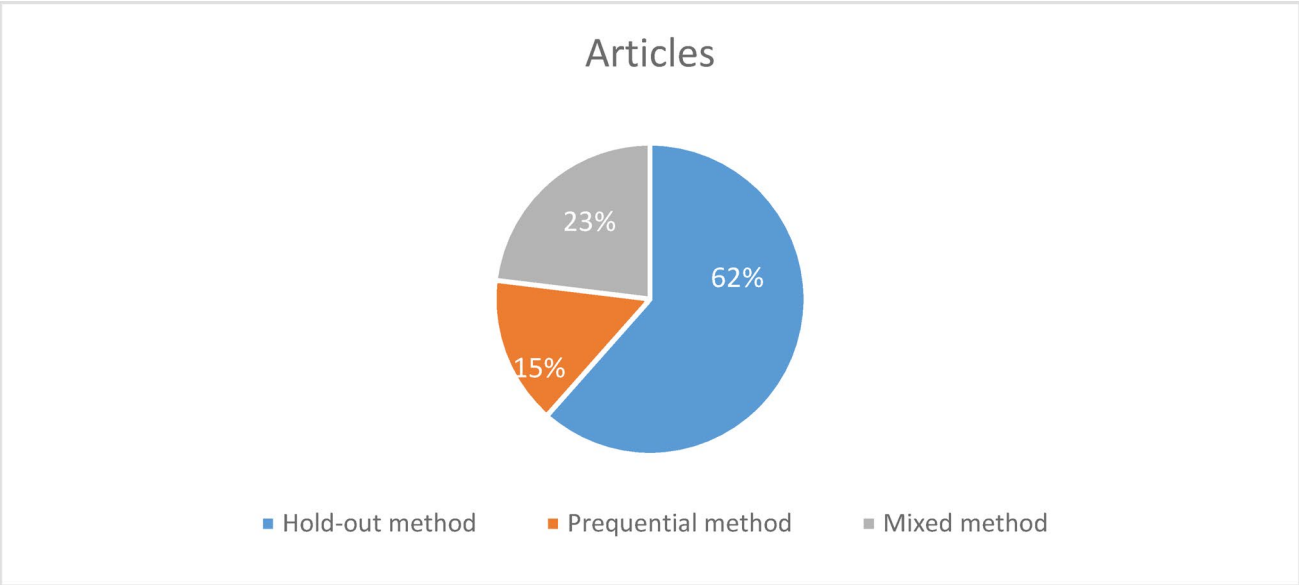


Fig. 2 Summary of validation methods

Recent developments in the fields of network analysis and anomaly detection shown tremendous evolution in modern cyber-attacks in terms of attack scope, techniques, damages and landscape. The set of articles reviewed has revealed the need for concept drift analysis for effective cyber-attack detection especially in a non-stationary network environment where data arrives in streams and subject to change in data distribution. The goal of concept drift learning in anomaly detection model is to ensure effective cyber-attack detection while preventing the model from deteriorating accuracy over time [33, 44, 45]. Consequently, several research works have been conducted and different concept drift learning techniques have been successfully implemented in network anomaly detection models. Therefore, a review of some of these articles are presented next.

Shao, Yuan and Wang [14] implemented adaptive window (ADWIN) for concept drift detection and detected network anomalous by comparing the performance of three machine learning algorithms, namely, Naive Bayes (NB), Hoeffding adaptive tree (HAT) and Adaptive random forest (ARF) in detecting Botnet attack using a self-generated anomaly detection dataset. The experimental results showed an increase in the performance of ARF including 0.9995 accuracy, 0.9990 Kappa statistics (K), and 0.9991 Kappa mean. These results show the effectiveness of concept drift approach to anomaly detection. A novel concept drift approach involving drift detection and adaptation methods known as optimized adaptive and sliding windowing (OASW) was presented in [2].

Concept detection process was handled by sliding window method and drift adaptation process was handled by particle swarm optimization (PSO) based adaptive window respectively. Anomaly detection process was performed by light gradient boosting machine (LightGBM) algorithm. The evaluation on two public IoT cyber-security datasets: IoTID20 and NSL-KDD in online and offline showed a significant improvement in anomaly detection in term of accuracy, precision, recall, F1 measure and average test time. This technique further improves the performance of its model by employing adaptive window-based PSO for handling or automatically adjusting the model to concept drift phenomenon, thereby, ensuring a consistent and accurate performance of the model.

More so, many studies have leveraged artificial neural network (ANN) and deep learning (DNN) with concept drift techniques to improve anomaly detection techniques. For instance, Xu et al., [15] proposed a concept drift adaptive technique to improve anomaly detection in IoT environment. The authors employed stratified sampling technique with adaptive weighted timestamp to handle drift detection and an improved Long Short-Term Memory (I-LSTM) neural network to detect anomaly in IoT network and benefitted from its forgetive mechanisms. The model was evaluated on a self-generated smart home emulated environment with seven attack types including denial of service (DoS) attack, malicious scan, malicious control, malicious operation, spying, data probing, and wrong setting categories. Following the parameter tuning process, the study employed RMSprop optimizer and Softlink activation and achieved 98% accuracy. A novel deep transfer learning-based IDS was proposed by [19] to detect network intrusion in IoT networks by employing residual neural network known as (P-ResNet). The evaluation of the proposed model showed its effectiveness in detecting intrusion in IoT networks.

Similarly, a concept drift technique featuring dynamic sliding window based on residual projection was proposed by [6] using convolution neural network (CNN) and Long Short-Term Memory neural network to detect anomalies in BoT-IoT dataset. The model was further evaluated on other non cyber security-based datasets including datasets SEA Concept and UG-2 C-5D. It was reported that the model achieved a significant improved performance on all the datasets used for the evaluation.

In a study conducted by Abdel-Wahab [46] towards developing anomaly detection system with the capability to work in a dynamic environment, detect concept drift and adapt to changes in input and output features. The author proposed a drift detection technique using Principal Component Analysis (PCA) method and online outlier detector, and employed online deep neural network based on hedge weighting mechanisms to adapt the model to new data. The evaluation on IoT traffic-based dataset known as DS2OS indicates that online deep neural network outperformed the regular deep neural network by achieving minimum of accuracy of 96.8% and maximum of 100% accuracy on different attack types in the dataset. In another research work, Seth, Singh and Kaur-Chahal (2021) used CICIDS2018 dataset to confirm the performance of their drift-adaptive model. The model comprises of Adaptive Random Forest (ARF) classifier and Adaptive Window (ADWIN) to detect change in the data stream and detect both known and unknown network intrusion attacks. The result from their experiment shows that the proposed model achieved overall accuracy and recall of 99.5% and 99.8% respectively.

Ensemble learning is another ML technique that is gaining wide adoption in recent time due to its capability to combine predictive ability of multiple algorithm to form a robust model [51]. Ensemble model is usually implemented to detect and adapt model to the occurrence of concept drift. In a recent study, an ensemble of two drift detection methods comprises of adaptive windowing (ADWIN) and drift detection method (DDM) was proposed with adaptive random forest (ARF) and streaming random patches (SRP) by [4] to construct four base learners following an initial subset sample selection using k-means cluster sampling method. The four base learners (ARF-ADWIN, ARF-DDM, SRP-ADWIN, and SRP-DDM) are weighted according to their real-time performance and integrated to construct a robust anomaly detection ensemble model with improved drift known as PWPAE. The evaluation of the proposed model on two datasets: IoTID20 and CICIDS2017 based on accuracy, precision, recall, and F1-score, and execution time showed a significant improvement in anomaly detection. Jain and Kaur [47] handled concept drift on NSL-KDD, CIDDS 2017 and Testbed generated datasets using a sliding window-based K-means clustering algorithm. By using ensemble of random forest (RF) and logistic regression (LR) for anomaly detection, an accuracy of 93%, 98% and 97% was achieved on NSL-KDD, CIDDS and Testbed datasets respectively. Wu et al., [48] proposed an ensemble framework with adaptive method called adaptive exponentially weighted average Ensemble (AEWAE). Ensemble of four learners was formed using ARF-ADWIN, ARF-EDDM, OPA and KNN-ADWIN. The framework involves data preprocessing, base model learning and online ensemble. The evaluation results on CICIDS2017 and IoTID20 datasets show that the framework achieved over 99% accuracy and precision on the two datasets with minimal latency.

Moreover, Wu et al., [49] proposed an incremental ensemble based on Relevance Vector Machine (RVM) technique to detect network intrusion. The method used a new Overall Misclassification Probability Voting Weight Value (OMPW) to dynamically update the ensemble model. The model achieved balance between the accuracy, robustness and resource consumption. The model was evaluated on KDD CUP 99 dataset and CICIDS2017 dataset. When compared to the benchmarked methods, the proposed method (DEIL-RVM) outperformed other techniques with over 98% detection rate on training data and 99% on the incremental sets. An ensemble bagging technique was proposed by [50] to provide a real-time anomaly detection in IoT. The study proposed an ensemble bagging technique based on deep learning, and evaluation on three network datasets, namely NSL-KDD, KDD CUP 99 and Kyoto 2006 + showed that the proposed model IbEC neural network achieved over 99% accuracy on NSL-KDD and KDD CUP 99 datasets and over 98% on Kyoto 2006 + dataset. An adaptive bagging technique based on internal weight update classifier scheme was proposed by [51]. The proposed method does not update all the classifiers in each iteration, rather, it updates only the worst performing classifiers based on the calculated weight. The evaluation on KDD CUP 99 showed that the proposed adaptive models (WUC and WUCP) recorded significant improvements over the base classifiers and bagging techniques. Martindale, Ismail and Talbert [35] proposed a heterogeneous ensemble intrusion detection technique to detect network intrusion or anomaly and respond to concept drift. The results using KDDCup99 dataset showed that adaptive random forest with Hoeffding Trees performed better than the Hoeffding Adaptive Tree (HAT) model.

Moreover, with the systematic review, we were able to extract meaningful information required to answer the set objects for this study. Therefore, the next sections present the structured information from the reviewed articles.

Concept Drift Detection Methods for Anomaly Detection Model

Concept drift detection methods include various algorithms for detecting concept drift in network data stream. The systematic review revealed some existing concept drift detection methods and categorized them into sampling-based technique, window-based technique and performance-based technique as depicted in Table 4.

- Sampling-based technique: uses a sampling method to select an appropriate portion of data samples for the purpose of identifying changes in data distribution based on time interval. For instance, in the study of Xu et al., [15], a stratified sampling technique with adaptive weight based on time interval was employed to detect concept drift in the IoT network data stream. The stratified sampling technique was employed to correct the imbalance data introduced by change in data distribution and adaptive weights method was introduced to improve the conventional stratified sampling method and emphasizes the importance of each time internal in detecting the existence of concept drift in data distribution. Similarly, K-means clustering was implemented with sliding window by [47] to handle data imbalance and aid in the identification of drift in the data stream. Principal component analysis (PCA) was also implemented by [46] with outlier detector to detect the occurrence of concept drift in the data stream. Generally, sampling-based technique works with the principle that the recent data stream has higher weights than the previous one. It also emphasizes the importance of sampling method for data balance and identification of change in data distribution in a data stream.
- Window-based drift detection techniques: uses sliding window to select the new data samples and discard the old ones. The advantages of this method are; it is very useful in detecting changes in the data stream, it improves the learning time, and reduces the storage space. A sliding windowing with fixed length is a drift detection algorithm that works by dropping the old data instances for a more recent data instances based on some pre-set rules [31, 47]. However, using a traditional sliding window with a fixed length is mostly considered to be difficult to implement because it problematic to fix appropriate window length for a non-stationary and continuous data that is characterized with dynamic behavior [52]. Hence, Adaptive Window (ADWIN) was implemented by [14, 39]. ADWIN is an advancement of a sliding window method with the capability of dynamically adjusting the length of its window with a consistent maximal length based on the assumption that the average value inside the window has not changed.
  Moreover, sometime it may not be efficient to use average value to determine the window length for detecting changes in data stream. One of the more efficient ways of improving the performance of Adaptive window-based drift detection is windowing-set optimization. Yang and Shami [2] proposed a concept drift detection method known as "Optimized Adaptive and Sliding Windowing (OASW)". This approach is equipped with particle swamp optimization (PSO) that automatically tune its hyper-parameters to adjust to the nature of any datasets. It also ensures that models are only updated only when it is necessary with less complexity.
- Performance-based drift detection technique: This method detects concept drift by monitoring degradation rate in the performance of the model. Two common examples of performance-based detection method according to this study are Drift Detection Method (DDM) and Early Drift Detection Method (EDDM). Yang, Manias and Shami [4] used DDM for drift detection and achieved a significant performance in detecting drift in the data stream.

Table 4 further presents the various concept drift detection methods implemented in the anomaly detection models for IoT networks. The table also categorizes the identified methods into three classes including stratified sampling method, window-based method and performance-based method.

Concept Drift Adaptive Techniques and Learning Approaches

One of the most important advantages of data stream learning is the ability to adapt machine learning techniques to the changes in data distribution occasioned by the concept and prevents the intelligence of predicting models becoming obsolete.

According to [2], there are three major methods for adapting stream learning based models to concept drift. The first one is the incremental learning which is used to partially update the learning model when a drift is detected. Other two are adaptive learning and ensemble learning. However, this study identified two trending techniques for anomaly detection methods: adaptive learning technique and ensemble learning technique as presented in Table 5:

- Adaptive learning techniques: adaptive learning method is commonly used to ensure that the learning model is

retrained or altered on a new/updated dataset after drift is detected. This study reveals that adaptive learning techniques was successfully implemented in many studies such as [14, 15, 19, 39, 46]. For instance, an adaptive learning based on Hoeffding adaptive tree (HAT) and improved Long Short-Term Memory (I-LSTM) neural network were proposed by [14]. In their study, HAT was used to alter the structure of the LSTM network and adjust the model to new data. The implementation of concept drift adaptive techniques has also improved the detection of attacks in the IoT network. Qiao, Novikov & Blech [6] detects concept drift using dynamic sliding window method known as dynamic residual projection method (DRPM) and adapt the separately used classifiers Convolution neural network (CNN) and Long Short-term memory (LSTM) to the drift while detecting botnet attacks in IoT. The technique was able to identify concept drift and improved the detection of botnet and prove the effectiveness of the model when implemented in a dynamic IoT environment.

- Ensemble learning techniques: this method provides a mechanism to combine and train multiple base learners with drift adaptive and anomaly detection capabilities. Recently, the implementation of concept drift adaptive method in an ensemble learning form is increasing. For instance [2], proposed an ensemble-based light gradient boosting machine (LightGBM) algorithm for detecting concept drift and improve detection of attacks. The method consists of exclusive feature bundling (EFB) and gradient-based one-side sampling (GOSS). The GOSS ensures that the time and memory usage of the model is reduced significantly, while FEB provide efficient feature selection for the learning models. Yang, Manias and Shami [4] combined two base learners to form ensemble of four different models including ARF-ADWIN, SRP-ADWIN, ARF-DDM and SRP-DDM. Another ensemble model was proposed by [47]. In their experiment, Random forest (RF) was combined with Logistic Regression (LR) as level 0 and level 1 base learner and later fed into Support Vector Machine (SVM) who serves as base 2 learner for detecting network-based attacks.

Evaluation Datasets and Cyber-Attacks

Anomaly detection model refers to machine learning based model for cyber-attacks detections. Therefore, machine learning technique placed high priority on the availability of evaluation datasets for training and testing of ML algorithms. Also, the effectiveness of anomaly detection method depends highly on the quality and recency of dataset to include attributes of new cyber-attacks. Similarly, data stream learning requires datasets with concept drift instances to be able to train algorithms on how to learn about different types of concept drifts for accurate detection and prediction. This section presents the summary of the various evaluation datasets used to train and test the concept drift-based anomaly detection models in the review articles and also presented their corresponding cyber-attacks. Table 6 presents the cyber-security-based datasets used in the selected studies to evaluate the impact of concept drift phenomenon on anomaly detection models in IoT networks. It also highlights the category of each dataset including self-generated dataset or public security dataset. The self-generated datasets are generated by the authors by setting up a testbed that emulates IoT environment. On the other hand, public cyber-security dataset are real network datasets that are publicly available for model training and testing. However, it is observed that the type of concept drift (real or virtual) as well as the duration the drift (abrupt, gradual, incremental) could not be accounted for in the used real-world datasets. Unlike in the synthetic datasets where it is possible to inject concept drift to a particular point to ascertain the nature and duration of concept drift.

This study reveals common evaluation dataset for concept drift based anomaly detection to include: smart home dataset (Xu et al., 2020); botnet dataset [14]; IoTID20 [3, 4]; NSL-KDD [3, 47]; CICIDS2017 [4]; Bot-IoT dataset [6]; Distributed Smart Space Orchestration System (DS2OS) [46]; Generated Testbed Dataset [47].

Evaluation Methods

Evaluation method plays a significant role in assessing the effectiveness and overall performance of a predictive model. The evaluation methods in this context cover the model validation methods and performance measures. Table 7 presents the evaluation strategies as extracted from some of the reviewed articles. As presented in Table 7, the review shows that there are three common validation methods for data stream-based anomaly detection models: the hold-out method, prequential and mixed evaluation method. The statistics of the articles with these methods are further indicated in Fig. 2:

Model Validation Methods

- Hold-out method: It is an evaluation method where small portion or percent of the dataset is reserved for training purpose and the remaining percent is dedicated

for testing. In most cases 10% of the dataset is used for training and the remaining 90% is allocated for online testing. According to the review, hold-out method is the most employed model validation method with 62%. This is probably because it can be used for both offline and online model evaluation purposes as observed from the studies [6, 15, 47, 53].

- Prequential method: This is an online based validation method for data stream mining-based model. It allows specific instance to be used initially for testing and then for training. This method is also referred to as an interleaved test-then-train validation method. In most cases, the prequential method is commonly used for streaming data or time-series based model where the chronological order of the data needs to be preserved and ensures that no future data is used for current prediction, but introduces new data that ensure model is refined and updated continually [48]. This method was successfully employed in some of the reviewed articles such as [14, 35] and it accounted for 15% of the analyzed reviewed articles.
- Mixed method: The mixed evaluation method involves implementation of both hold-out and prequential methods for validation purpose. This method seems to becoming popular among the researchers in the field in the future. This method accounted for 23% of the analyzed and synthesized articles for model evaluation purposed. As can be seeing from the studies of [3, 4, 48], the method is mostly employed by implementing hold-out for offline learning, the prequential for online learning.

Model Performance Evaluation Methods

Machine learning techniques rely on several mathematical and statistical based metrics to evaluate the performance of the developed model. The performance evaluation plays a key role in determining the efficiency and effectives. As shown in Table 7, the common evaluation metrics are accuracy, precision, recall, f1-score, execution time (delay), kappa statistics and confusion matrices. The "tick" and "hyphen" signs indicate the used and unused metrics respectively in the reviewed articles. Therefore, out of the aforementioned metrics, accuracy, recall, precision and f1-scores are the most metrics in the reviewed articles. Meanwhile, accuracy appeared as the most used metric for data stream or drift-based anomaly detection methods, while kappa statistics revealed as least used metrics. Although, accuracy is the most popular in classification method, it is always not only the de-facto metric to ascertain the performance of models. Other metrics, such as confusion metrices, recall, F1-score, and precision, are equally very useful in determining effectiveness of a model, most especially in the cases of data imbalance [15, 46, 47].

Number of Evaluation Datasets

Studies have shown that the ability of a machine learning model to achieve higher or reasonable performance on different dataset also goes a long way in determining it effectiveness. One of the revelations of this study is number of datasets used in evaluating the anomaly methods with concept drift approach. As indicated in Table 7, the review shows that most of the existing anomaly models with data stream learning used more than one dataset to evaluate the performance of the proposed models. This could be a trend in this field to ensure the effectiveness of the models when deployed for real-time cyber-attack detection in a dynamic network environment.

Discussions and Implications for Research Direction

This section focuses on discussion of findings from the systematic review with a view to answer questions bordered on identifying the existing concept drift detection methods, the techniques and strategies for adapting anomaly detection mode to concept drift phenomenon, the datasets employed for model training and testing in their inherent cyber-attackers, as well as the general and specific methods for validating and evaluating the model. This section also discusses the implication of the finding for research directions.

Concept Drift Detection Methods for Anomaly Detection Model in IoT Networks

Concept drift is an inevitable and unforeseen data stream phenomenon that depletes the accuracy of predictive models if not properly addressed. The results of the systematic review revealed some existing drift detection methods, which are often referred to as drift detectors. From the review, the existing drift detectors can be categorized into three in term of the techniques employed by the algorithms to detect changes in data distribution. The result of the systematic review presented in Sect. 4.1 shows that the first category of the existing concept drift detection method (that is, sampling-based drift detection technique) uses statistical sampling method such as stratified sampling technique [15], K-means clustering [47], PCA [46] to identify the occurrence of drift in data stream. The advantage of this method lies in its ability to select the most informative and representative portion of data to balance sample data, assigning

appropriate weight to the selected data for easy identification of change in data distribution that causes concept drift.

Window-based drift detectors are the second category of the trending concept detection methods for IoT anomaly detections. This method employs sliding window algorithm to detect the occurrence of concept drift in IoT data stream. Some of the advantages of the sliding window method include simplicity, improved learning time and effective forget mechanism. A sliding window algorithm replaces old concepts with new concepts by discarding the old data samples with new and most recent samples. Sliding window is usually implemented with either a fixed length or varied length. In the fixed length mechanism, a preset rule is employed to determine the length of the window [31, 47]. This method is simple and fast to implement for a fewer dynamic data samples, but very difficult for a large and highly dynamic data stream environments. Therefore, adaptive windowing method provides a more robust method with varied and automatically adjustable window lengths. This method is gaining wide adoption and has been used successfully in many studies such as [14, 39]. Nevertheless, the nature of concept drift some times makes automatic determination of window length not efficient enough. In such cases, ADWIN methods are equiped with optimization mechanisms as seeing in the studies of [3] with particle swarm optimization. The third category of concept drift detection method is called performance-based drift detection technique. Examples of this method include drift detection method (DDM) and early drift detection method (EDDM) as implemented in [4]. So, the dynamic nature of IoT demands that more efficient and optimized concept drift detection methods are needed to cope with the ever evolving and voluminous nature of its data for effective detection.

Data Stream Learning for IoT Anomaly and Cyber-Attacks Detection

IoT networks are dynamic, and consist of non-stationary and voluminous data that are continuously generated by billions of IoT devices for real-time processing and smart services. Meanwhile, IoT devices and servers require adequate memory storage for real-time processing. However, with the inherent memory constraints of IoT devices, stream learning appears to be an ideal solution for this problem. Therefore, the findings from our systematic review show that there is need for increasing adoption of stream learning for IoT anomaly detection systems. Unlike the traditional batch learning where anomaly models are trained on static data ones, stream learning allows classifiers to be trained and retrained online and learn in real-time. Stream learning handles the memory problems using some strategies to classify arriving data stream into current and old data, the training and retraining of the model are performed using current data, while technique such as forgetting mechanism could be used to discard to old data, thereby free up spaces for real-time processing.

The Need for Concept Drift Approaches for IoT Networks

Concept drift is a stream learning problem that depletes the performance of stream learning-based anomaly detection models. With the ubiquitous nature of IoT devices and increasing adoption for various services and applications, more devices with different user characteristics and motives are joining the IoT network ecosystem, thereby making the prediction of IoT networks and users' behaviors becoming more difficult. Therefore, deploying anomaly detection model without the capability of detecting change in data distribution, known as concept drift, will make the models to be ineffective in detecting both known and unknown cyber-attacks, while ensuring a secure learning for ML classifiers. For instance, since adversarial attacks involve injection of noise and poisoned data to input dataset to cause changes in data distribution, integrating concept drift detectors with adversarial attacks defense system is suggested by this study in future research efforts. With powerful drift detector, malicious changes in data distribution due to adversarial injections could be detected and handled earlier to prevent it from impacting the performance of adversarial attacks detection model or others. Hence, this study calls for more concept drift approaches and revealed the need for incorporating concept drift approaches for dynamic network environments such as IoTs.

Support for More and Better Concept Drift Detection Methods

Our systematic review has shown that several concept drift detection methods have been developed, evaluated and proposed in the literature. However, more techniques are still required to cope with the dynamic and unpredictable changes in network characteristics as a result of unstable traffic volumes, network patterns and security attack signatures. Therefore, it becomes imperative to retrain the models constantly to address the changes occasioned by the concept drift phenomenon. Similarly, better concept drift methods are still needed to address some of the issues in the existing drift detection methods such as delay detection, false detection, inaccurate detection, and computation overhead.

Many studies have reported the robustness of deep neural network-based models in handling data related issues such as data imbalance, high complexity, and dimensionality in IoT network environments. Deep learning methods like CNN and LSTM [6] I-LSTM [15], P-RestNet [19],

and Autoencoder with Gare Recurrent Unit (GRU) [54] have been successfully implemented with overwhelming improved and efficient performance with regard to these issues. More importantly, these studies have equally benefitted from optimization algorithms for hyperparameter and computational overhead reduction. For instance [15], enhanced the performance of I-LSTM with RMSprop optimizer and softlink activator, Honey Badger Algorithm (HBA) was employed in [54], and PSO was implemented in [2].

Concept Drift Detection Approaches and Computational Overhead

The review shows an increase in the adoption of adaptive and ensemble learning techniques for concept drift-based anomaly detection model in IoT networks. However, with resource constraint nature of IoT systems, these techniques require more resources, such as processors, memory, and bandwidth, to implement and mostly lead to computational overhead [3, 4]. Although. recent studies [4, 39, 47, 48] have demonstrated how adaptive and ensemble learning techniques are implemented in IoT with no or less computational overhead. For instance, adaptive techniques have been implemented with edge or fog computing in [55, 56] where narrow band-IoT (NB-IoT) and Adaptive Heuristic Edge assisted Fog Computing design (AHE-FCD) were proposed respectively, lightweight algorithms like LightGBM was implemented in [2], forgetfulness mechanisms in [15] where improved Long Short-Term Memory (I-LSTM) neural network was employed, and incremental learning approach in [49] where Relevance Vector Machine (RVM) where necessary parameters are updated only when drift is detected [14, 15, 19, 39, 46]. Similarly, studies have shown that strategies such as combination of lightweight models, use of hierarchical methods, parallel processing and pruning methods have been used successfully to handle ensemble-based models in IoT environment [2, 4, 47]. Nevertheless, with respect to the sophistication of modern cyber-attacks on cyber-physical systems, more efficient and intelligent coping strategies are required to develop effective and efficient adaptive or ensemble-based models to provide cyber-resilient solutions for IoT systems. In a recent study by [54], the author proposed a hybrid model based on Autoencoder and Gated Recurrent Unit (GRU) to enhanced detection of cyber attack threats in IoT environment. This model was further improved by Honey Badger Algorithm (HBA) to optimize the parameter of the GRU for robust cyber-attack detection. This study underscores the importance of hybridization and ensemble techniques with parameter optimization strategy to achieve an effective and efficient model.

Strategies and Approaches for Adapting Anomaly Detection Model with Concept Drift

Concept drift adaptation is an essential phase in concept drift learning and analysis. With drift adaptation technique, anomaly detection model will be able to adjust itself by implementing a strategy that will adapt the anomaly detection model to changes in data distribution and ensure that the classification or prediction output remain accurate. The essence of the adaptation phase is to complement the limitation of drift detection methods such as delay or false drift detection [38]. The two common adaptation model according to the review are: adaptive learning technique and ensemble learning technique.

Adaptive learning techniques is an extension of incremental learning and it is usually implemented based on two configurations [14]. In the first configuration, the learning model is retrained as a result of drift detection, and such configuration can be found adaptive classifier such as Hoeffding Adaptive Tree (HAT). In the other configuration, a temporary or based learning model is configured for classification alongside a corresponding background model, and upon detection of concept drift, the affected base learning model will be replaced with a background learning model, and such configuration can be found adaptive classifier such as Adaptive Random Forest (ARF). Meanwhile, this study revealed that many adaptive learning techniques have been implemented like HAT [14], I-LSTM [15], ARF [39], CNN and LSTM [6], Online Deep neural network [46] as shown on Table 7. Overall, the advantage of adaptive learning strategy is it ability to adapt anomaly model to changes in data distribution automatically and improve online learning accuracy. Most of the existing adaptive learning models are implemented as a single learning classifier, and are sometimes not effective or robust enough to cope with the dynamic and sophistication of some cyber-attacks such as distributed denial of service (DDoS) [3, 36] and adversarial attacks [40].

Therefore, to compensate for the limitation of single learning adaptive classifier, this study identified some studies that have implemented ensemble learning technique to improve both the adaptation strategy and online detection accuracy of anomaly detection model in IoT networks. For instance, as presented in Table 7, ensemble LightGBM model was proposed by [3], ensemble of four classifiers including ARF-ADWIN + ARF-DDM + SRP-ADWIN + SRP-DDM was proposed by [4], ensemble of three classifiers including LR + RF + SVM was proposed by [47] among others. The review shows that the implementation of ensemble learning actually achieved improved performance over the single classifiers in their various studies, however, factors that determines the actual number of base classifiers

that could be combined was not considered as there is need to strike balance between the model accuracy and computation overhead including the learning time.

Support for More and Better Drift Adaptation Methods

This study revealed that adaptive and ensemble learning techniques are the trending methods for adapting network anomaly detection models to changes. However, with the dynamic and sophistication of IoT data and security attack signature, more and better techniques are still needed to constantly retraining or adapt the ML model to changes without causing additional computational overhead to IoT system. The review shows a trend in the development of ensemble learning techniques to compensate for ineffectiveness of single adaptive ML classifiers by combining multiple classifiers for better performance. But, most of the existing ensemble models have not established the parameters that determine the appropriate number of classifiers that can be combined to build an effective ensemble network anomaly detection model with adaptive mechanisms with unnecessary computational overhead while maintaining optimal accuracy. Thus, this remain an open research direction to ensure a safer IoT ecosystem.

Security Datasets and Implications for Concept Drift Evaluation

As presented in Sect. 4.3, several security datasets have been used to evaluate the various concept drift-based anomaly detection models in the literature. The proliferation of the evaluation datasets is as a result of rapid evolution of cyber-attacks and dynamic nature of IoT networks. As shown in Table 6, these evaluation datasets are either self-generated or synthetically generated by authors in a testbed environment [15, 47] or a real world public security dataset such as NSL-KDD [3], IoTID20); IoTID20 [3, 4]; NSL-KDD [3, 47]; CICIDS2017 [4]; Bot-IoT dataset [6]. This study revealed that each of the datasets contains concept drift in their network traffic and the analysis shows the influence of concept drift detection and adaptation. The analysis therefore demonstrates the importance of incorporating concept drift approaches in a model for a dynamic environment like IoT networks for effective cyber-attack detection. More so, we also highlighted the attack types in each of the identified datasets, such as DoS, botnet, Probe and User to Root (U2R) Probe and User to Root (U2R), web attacks, Brute Force, Slowloris, Slow HTTP Body and Slow HTTP Read, Network Scan, Spying, and Wrong Setup, Mirai botnet attack, MITM. More so, our study also shows that DoS, botnet, probing, and malicious scanning are the most prominent IoT cyber-attacks in most of the evaluation datasets in the primary studies. This finding may suggest the cyber-attack trend and the research direction as far as the IoT security is concern.

In addition, as observed in Sect. 4.3, most of the real-world datasets used for concept drift analysis could not be used to analyze the nature and duration of concept drift in the datasets. Several studies have shown the importance of using real-world dataset for machine learning model, but the use of real-world datasets with known concept type, location, and duration is still an open research in this domain. Therefore, this study sees this as a research gap and a future research direction.

Evaluation Methods for Stream Learning Model and Need for Standardization

This study revealed three common concept drift-based validation methods including hold-out, prequential and mixed method. As shown in Fig. 1, hold-out is the most used validation method with 62% of the reviewed articles. The mixed method validation revealed as the second most used method with 23%, while the prequential method reported as 15%. The hold-out validation method is widely employed but the prequential and mixed methods are gaining increasing adoption as they are developed for data stream learning or online learning. More so, our systematic review identified seven common performance metrics used for evaluating the performance of the proposed cyber-attack detection methods including accuracy, precision, recall, f1-score, execution time (delay), kappa statistics and confusion matrices. As shown in Table 7, accuracy, precision, recall, and f-score are the prominent metrics. This revealed the significance of these metrics for concept drift-based anomaly detection models. In addition, Sect. 4.4.3 presents the synthesis of the number of datasets used for evaluating the proposed models in the reviewed articles. As revealed in Table 7, the result of the analysis and synthesis of the articles show that most of the models were evaluated on more than one dataset. The review shows two or three datasets were used in most of the articles. This suggests a trend in the number of datasets for model valuation.

Therefore, the disparity in the evaluation methods in the existing studies as reveled by this study shows that it is difficult to compare and contrast results from various studies simply indicates the absence of standards metrics, evaluation and validation methods, training and testing datasets. This call for research efforts from researchers as well as standard promoting organizations such as Internet Engineering Task Force (IETF) to work towards standardizing the evaluation methods for effective and objective comparative analysis of the studies.

Summary of Trends IoT and Concept Drifts Methods

This study observed that there is an incremental growth in the adoption of concept drift-aware methods in the literature within the period covered. It observes a shift from the use of methods that only detect drifts such as [6, 46, 47] to the methods that integrate both concept drift detection and adaptation such as [2, 5, 14, 15, 48, 50] where techniques like ensemble, and adaptive learning are used. The review also recognizes the dominance of supervised-based methods in the field and observed a tilt towards the use of unsupervised methods in recent studies such as [4, 14, 47, 54] for handling scarcity of label data, high dimensionality and complexity of IoT data. Similarly, the study observed an increasing implementation of deep learning with optimization methods, thereby bridging the gap between building a robust model and maintaining online learning as in [2, 6, 19, 54]. More application focus solution is also springing up to target specific IoT network environment or domain such as IDS [39], smart grid [41], system log [53], healthcare [55, 56].

Conclusions

Internet of Things consist of millions of smart devices with voluminous amount of data that are continuously generated and processed for real-time smart services. The IoT traffics arrive in streams and requires the analysis and processing of data stream to be done online for real-time anomaly and cyber-attack detections. Concept drift is a phenomenon that occurs as a result of changes in data distribution and it has adverse effect on the accuracy of cyber-attack detection in a non-stationary IoT network environment. The concept drift causes deterioration in anomaly detection models, thereby making detection of cyber-attacks a difficult task in IoT environment. Therefore, this work conducted a systematic review of studies that have incorporated concept drift approaches for IoT networks anomalies detection between 2020 and 2023 from reputable academic journals and databases in order to keep track of the recent techniques in the areas of concept drift detection models and classifiers for stream learning. Meanwhile, this study has investigated the concept drift approaches with a focus on drift detection methods, strategies for adapting anomaly models to concept drift, prominent datasets and methods for evaluating the models. The study has revealed the trending concept drift detection such as ADWIN, K-means based sliding window, PCA, DDM and EDDM with their categories. It also revealed adaptive and ensemble methods as the common classes for drift adaptation in the selected articles. The review confirms the presence of concept drift in most of the evaluation datasets, and the common evaluation and validation methods were identified.

Based on the findings from the various article reviewed, more research efforts are required in the area of stream learning.

For more effective data analysis, future research direction could focus on developing methods that encourage hybridization of concept drift and classifier to provide continuous online leaning with effective drift detection and adaptation. Deployment of unsupervised method to handle label data deficiency and adapt to shift in data efficiently.

In term of cyber-attack detection, more novel methods are required to effectively identify and adapt anomaly detection models to concept drift phenomenon in IoT networks. This study shows that less research attention is being paid to the implementation of unsupervised and semi-supervised based concept drift detection approaches. For instance, out of the selected articles, only four studies have implemented unsupervised or clustering based approach such as K-means clustering [4, 15, 47], and Auto encoder (AE) algorithm for handling issues of class imbalance, absence of label data in real-world setting, and dimensionality reduction [54]. Given the effectiveness and application of unsupervised techniques to real life scenarios, availability and cost of acquiring labelled data, more research efforts in this direction are suggested. Federated learning with privacy preserving method may be employed to detect local drifts, adapt and aggregate changes, and preserve privacy IoT users. More research efforts on benchmarking evaluation dataset, metrics, and protocols are very important for the future of this research domain.

More so, this study revealed that DoS, botnet, probing, and malicious scanning are the most prominent IoT cyber-attacks in the popular evaluation datasets in the IoT network anomaly detection domain. More possible research directions could focus on developing more efficient techniques for detecting highly sophisticated IoT attacks such as DDoS. In term of performance evaluation metrics, more research efforts to identify the most effective and standard metrics for evaluating concept drift-based models. A framework to selecting real-world datasets for analyzing concept drift occurrence, nature of the drift, and time of occurrence is still required. The review does not consider other non-concept drift enabled techniques and only focus on solutions for IoT networks. With regards to the evaluation datasets, we only considered identifying the most common datasets and cyber-attack types for evaluation. These limitations may be investigated in future research to further enhance the design and development of machine learning based models.

Finally, we hope this paper will provide researchers with state-of-the-art knowledge on the recent research development on concept drift techniques for IoT anomaly detection

to support development of better and more effective concept drift enabled solutions. The implications of our findings called for more state-of-the-art researches to advance the security of IoT systems to achieve the long-time vision of smart computing for automatic service delivery (smart services).

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Declarations

Conflict of interest We do not have any competing interest.

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